

Fan-out benchmark - simplicio-prompt kernel on sistema-sindico

Date: 2026-05-28 Model: meta-llama/llama-3.1-8b-instruct temperature: 0.7 Tasks: 8 N values: 64, 200

Engine: kernel.subagent_runtime.SubagentRuntime (PyPI simplicio-prompt 1.7.0). For each (task, N) the kernel launches N real parallel LLM calls through LaneWorkerPool on the same prompt; each output is scored by real PHPUnit; the majority-vote outcome (sha256-mode of normalized code) is re-scored. Central question: does N=64 (sp default) help vs lower N? does N=200 help vs 64?

Headline - per N (aggregate across tasks)

N	Tasks	Per-attempt	Modal pass	Tokens	Cost (USD)	Avg elapsed
64 (default)	8	356/512 (69%)	5/8	675,616	\$0.0405	41.3s
200	8	1343/1600 (83%)	7/8	2,126,410	\$0.1276	39.2s

Per-task

Task	N	Per-attempt	Uniq	Modal	Tokens	Cost
password_strength	64	64/64 (100%)	10	P 33/64	65,398	\$0.0039
password_strength	200	200/200 (100%)	24	P 163/200	204,825	\$0.0123
password_require_symbol	64	12/64 (18%)	17	. 37/64	61,215	\$0.0037
password_require_symbol	200	166/200 (83%)	47	P 137/200	190,731	\$0.0114
env_get_int	64	53/64 (82%)	17	P 34/64	61,346	\$0.0037
env_get_int	200	180/200 (90%)	30	P 137/200	192,874	\$0.0116
env_get_bool	64	13/64 (20%)	28	. 33/64	61,039	\$0.0037
env_get_bool	200	155/200 (77%)	50	P 137/200	199,687	\$0.0120
admin_only_allowed_roles	64	64/64 (100%)	2	P 61/64	32,449	\$0.0019
admin_only_allowed_roles	200	200/200 (100%)	2	P 196/200	102,008	\$0.0061
rate_limit_bucket_key	64	64/64 (100%)	13	P 36/64	90,217	\$0.0054
rate_limit_bucket_key	200	199/200 (99%)	17	P 137/200	282,474	\$0.0169
base_repository_build_where	64	23/64 (35%)	15	. 33/64	157,915	\$0.0095
base_repository_build_where	200	45/200 (22%)	45	. 137/200	495,439	\$0.0297
router_has	64	63/64 (98%)	20	P 33/64	146,037	\$0.0088
router_has	200	198/200 (99%)	41	P 144/200	458,372	\$0.0275

Interpretation

Per-attempt pass = noise floor at temp=0.7 (single-call equivalent). Modal-vote = does picking the most frequent answer recover correctness? If N=200 doesn't beat N=64 on modal pass-rate, the sp default is the sweet spot. Cost scales linearly with N for tokens; wall-clock grows sub-linearly because calls run in parallel through LaneWorkerPool.